Experiment 14

14.1.1 Complex Number Addition using Class Algorithm :

Step 1: Start

Step 2: Define class Complex with three methods:

- **initComplex(self)**
- **sum(self, c1, c2)**
- **display(self)**

Step 3: Method initComplex(self)

3.1 **Read two integers from input**

3.2 **Assign first value to self.real**

3.3 **Assign second value to self.img**

Step 4: Method sum(self, c1, c2)

4.1 **Set self.real = c1.real + c2.real**

4.2 **Set self.img = c1.img + c2.img**

Step 5: Method display(self)

5.1 If self.img >= 0:

Print self.real + self.img i

5.2 Else:

Print self.real - |self.img| i

Step 6: Create three objects

6.1 **c1 = Complex()**

6.2 **c2 = Complex()**

6.3 **c3 = Complex()**

Step 7: Initialize c1 and c2

7.1 **Call c1.initComplex()**

Read and assign real and img for c1

7.2 **Call c2.initComplex()**

Read and assign real and img for c2

Step 8: Compute sum

8.1 **Call c3.sum(c1, c2)**

8.2 **c3.real = c1.real + c2.real**

8.3 **c3.img = c1.img + c2.img**

Step 9: Display result

9.1 **Call c3.display()**

9.2 **If c3.img >= 0:**

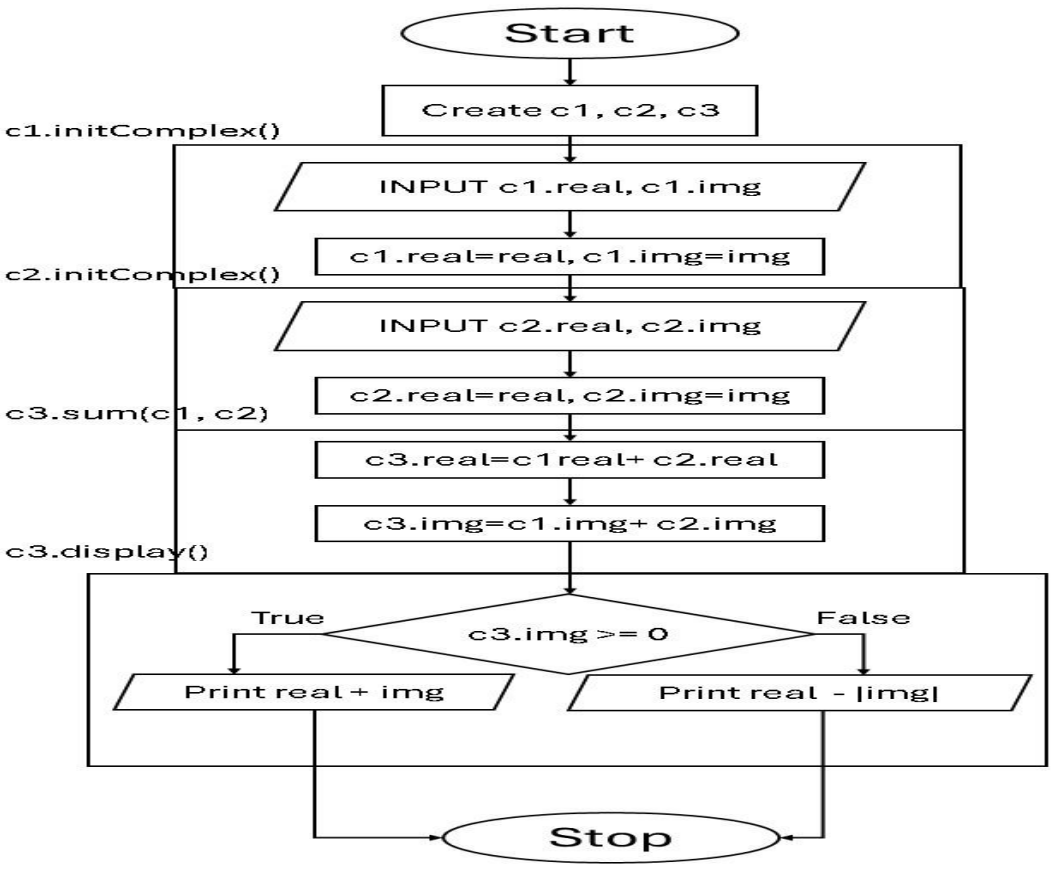
Print c3.real + c3.img i

9.3 Else:

Print c3.real - |c3.img| i

Step 10: Stop

Flowchart :



Code :

```
class Complex:
def
initComplex(self
):
self.real , self.img = map(int,input().split())

def sum(self,c1,c2):
self.real = c1.real + c2.real
self.img = c1.img + c2.img

def     display(self):     if
self.img          >=          0:
print(f"{self.real}          +
{self.img}i") else:
print(f"{self.real} - {abs(self.img)}i")
# Create three
instances c1 =
Complex() c2 =
Complex() c3 =
Complex()

# Initialize two complex
numbers c1.initComplex()
c2.initComplex()

# Compute and
display sum
c3.sum(c1, c2)
c3.display()
```

Execution :

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14.1.1. Complex Number Addition using Class 15:28

Write a Python program using a class named **Complex** to add two complex numbers.

Create a class **Complex** with the following methods:

- **initComplex()**: Read user input for the real and imaginary parts.
- **sum()**: Compute the sum of two complex numbers and store the result in the current instance.
- **display()**: Display the complex number in the format "<a> + <b>i" or "<a> - <b>i".

The program should create three instances of **Complex** class: two for input complex numbers and one for the result. Initialize both complex numbers, compute their sum, and display the result.

Input Format:

- First line contains two space-separated numbers: real and imaginary parts of first complex number
- Second line contains two space-separated numbers: real and imaginary parts of second complex number

Output Format:

- Print a single line showing the sum of the two complex numbers in the format "<a> + <b>i" or "<a> - <b>i"

Explorer

complexN... Submit

```
1  class Complex:
2      def initComplex(self):
3          self.real , self.image =
   map(int,input().split())
4
5      def sum(self, c1 , c2):
6          self.real = c1.real + c2.real
7          self.image = c1.image + c2.image
```

Debugger

Average time **0.009 s** 8.50 ms

Maximum time **0.014 s** 14.00 ms

4 out of 4 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 14 ms Debug

Expected output

3 4

2 5

5 + 9i

Actual output

3 4

2 5

Terminal Test cases

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